

Multiclass Classification with LightGBM

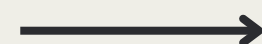
THREE WISE MONKE



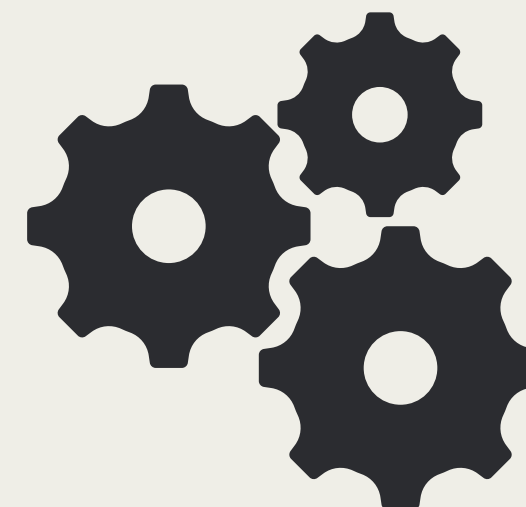
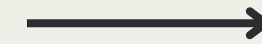
METODOLOGY



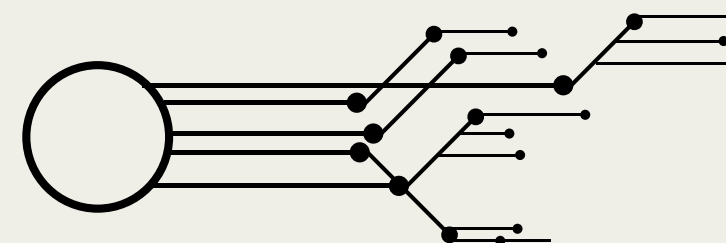
Exploratory Data Analysis



Data Preprocessing



Feature Engineering



Modelling



Validation

DATASET DESCRIPTION

Data Train

54480 rows and 28 columns

Data Test

13599 rows and 27 columns

Task

Cyber attack technique
classification

Metric Evaluation

$$F1\ Score = 2 \times \frac{recall \times precision}{recall + precision}$$

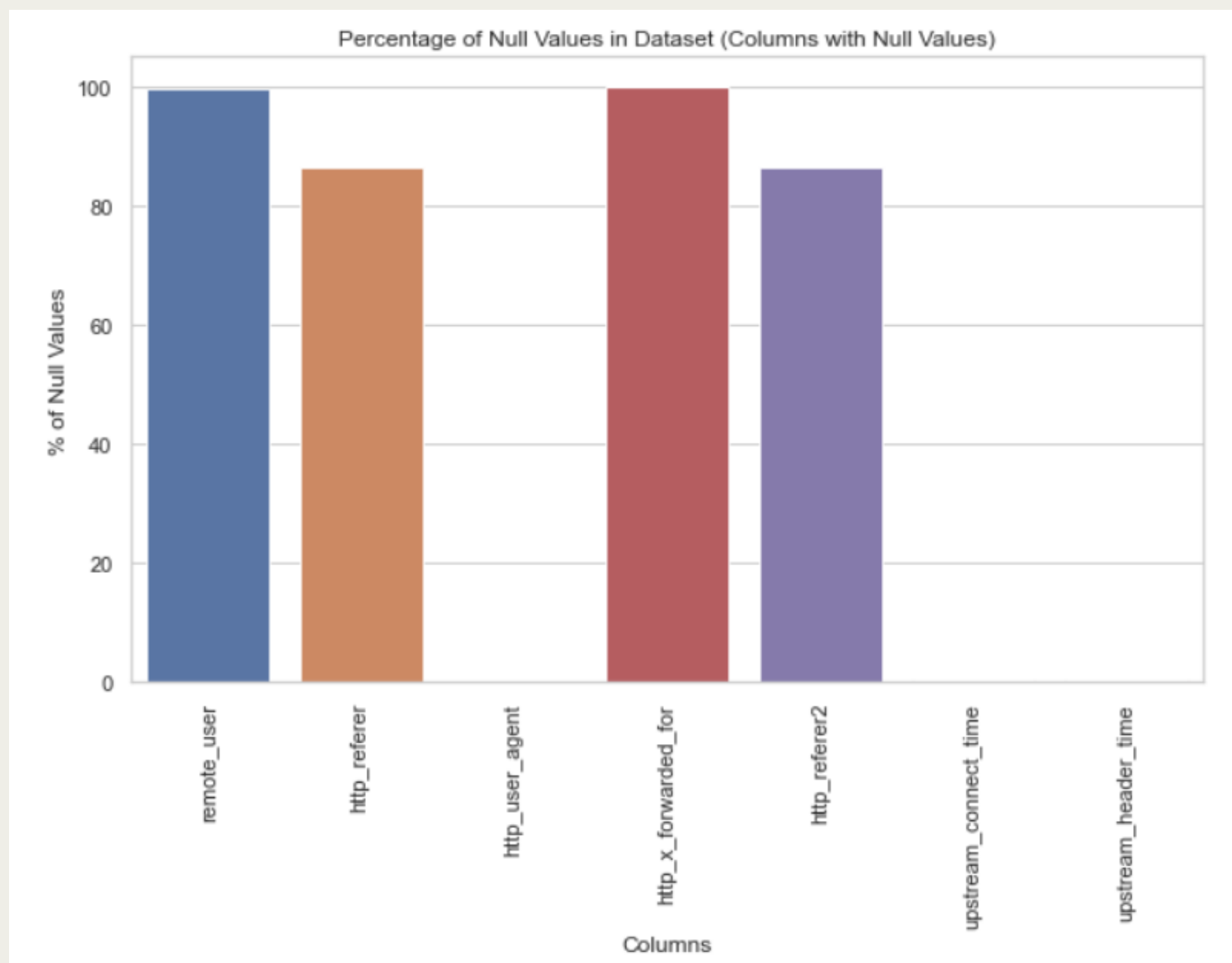
Label Meaning

["Credential Access"]: CMS (WordPress or Joomla) login attempt.
["Defense Evasion"]: Suspicious URL access.
["Impact"]: High amount of POST requests in a small period of time (likely bot).
["non-attack"]: non-attack.
["Initial Access"]: A web |attack returned code 200 (success).
["Reconnaissance"]: Multiple web server 400 error codes from same source ip.

Features

id	int64
timestamp	object
remote_addr	object
remote_user	object
hostname	object
time_local	object
request	object
status	int64
body_bytes_sent	int64
http_referer	object
http_user_agent	object
http_x_forwarded_for	object
time_iso8601	object
request_method	object
request_length	int64
bytes_sent	int64
http_referer2	object
request_time	float64
upstream_connect_time	object
upstream_header_time	object
upstream_response_time	object
upstream_addr	object
upstream_status	int64
connection	int64
connection_requests	int64
data.srcip	object
agent.ip	object
rule.mitre.tactic	object

NULL VALUES



Missing numerical train data:

upstream_header_time Null Values: 0.55% of the dataset (302)

upstream_connect_time Null Values: 0.46% of the dataset (251)

Missing categorical train data:

http_x_forwarded_for Null Values: 100.0% of the dataset (54477)

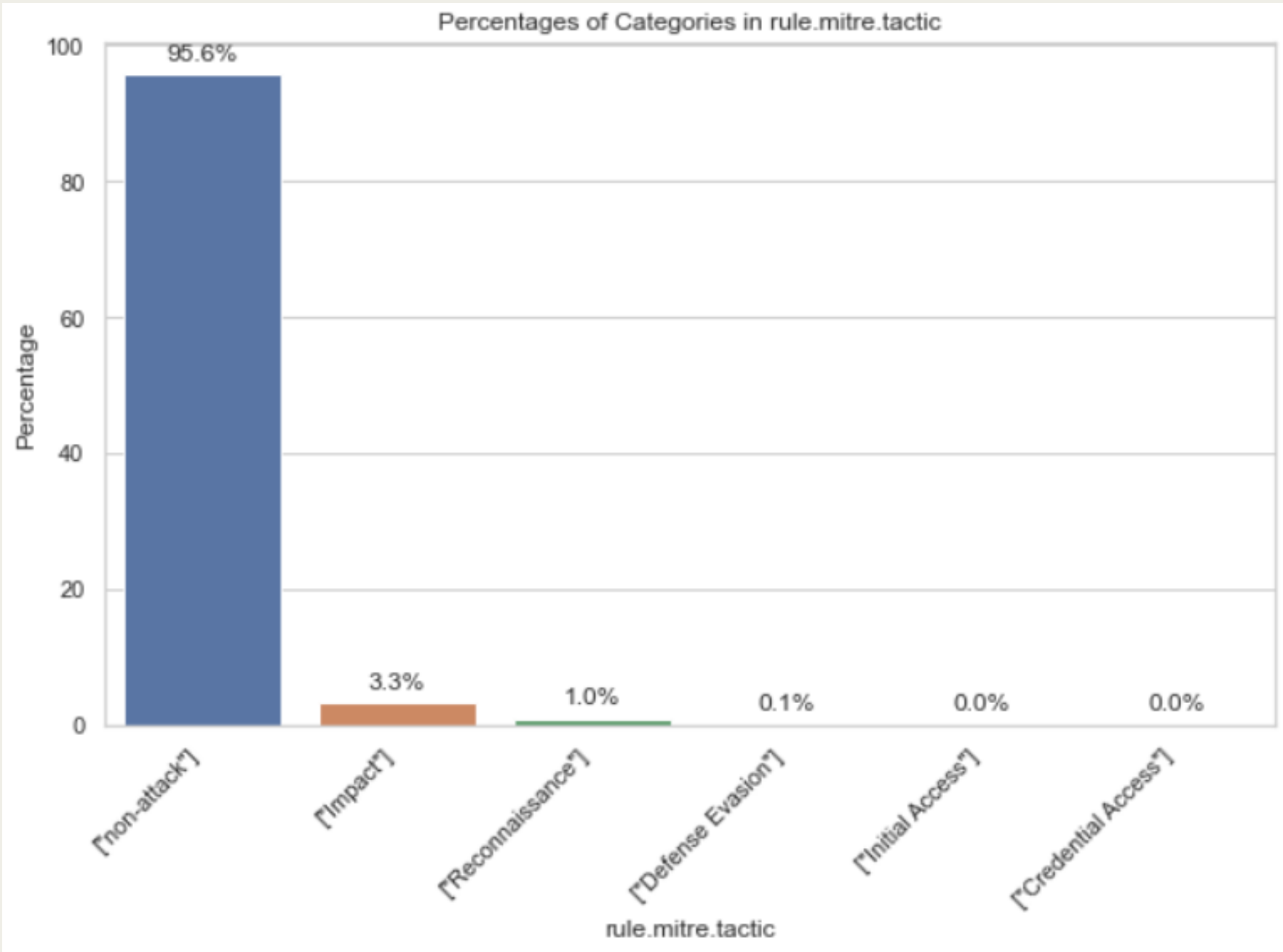
remote_user Null Values: 99.72% of the dataset (54328)

http_referer Null Values: 86.47% of the dataset (47110)

http_referer2 Null Values: 86.47% of the dataset (47110)

http_user_agent Null Values: 0.06% of the dataset (33)

TARGET DISTRIBUTION

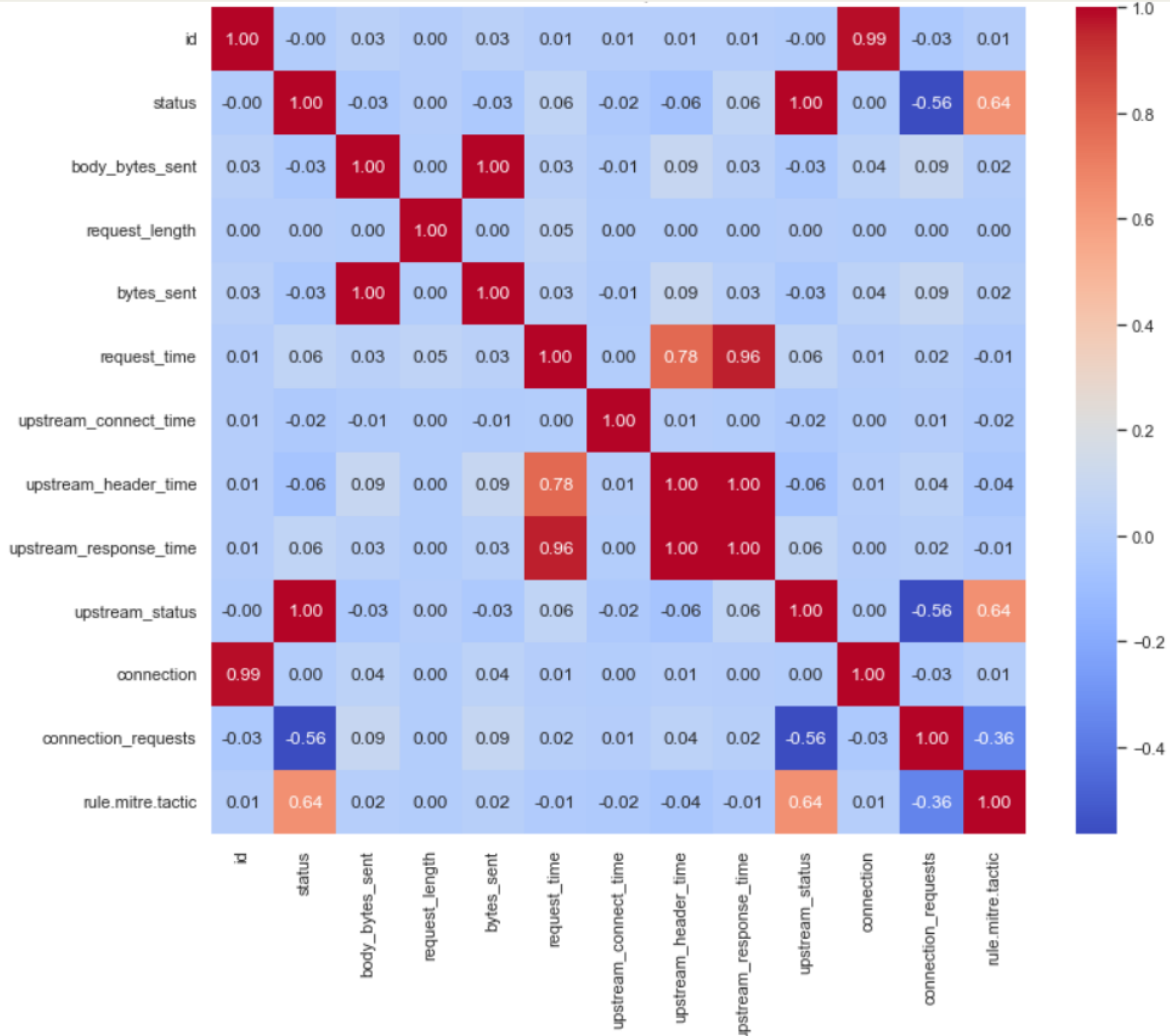


```
rule.mitre.tactic
["non-attack"]      52101
["Impact"]          1784
["Reconnaissance"]  561
["Defense Evasion"] 29
["Initial Access"]  4
["Credential Access"] 1
Name: count, dtype: int64
```

CLEANING & PREPROCESSING

Replace “-” with NaN	Reformatting Data Type
<pre>upstream_connect_time upstream_header_time - -</pre>	<pre>upstream_response_time object upstream_header_time object upstream_connect_time object</pre> <pre>upstream_response_time float64 upstream_header_time float64 upstream_connect_time float64</pre>
Simplify Multivalued Columns	Drop “Credential Access” Target
<pre>14 0.000, 0.001, 0.000 455 0.000, 0.001, 0.000 726 0.001, 0.000, 0.000 727 0.000, 0.000, 0.001 728 0.000, 0.000, 0.001 ... 42371 0.000, 0.001, 0.000 44034 0.016, 0.017 49931 3.005, 0.001 50480 0.000, 0.000, 0.001 50793 0.000, 0.001, 0.000 Name: upstream_response_time, Length: 166, dtype: object</pre> <pre>0 0.002 1 0.002 2 0.002 3 0.002 4 0.003 ... 54475 0.002 54476 0.001 54477 0.002 54478 0.002 54479 0.001 Name: upstream_response_time, Length: 54480, dtype: float64</pre>	<pre>rule.mitre.tactic ["non-attack"] 52101 ["Impact"] 1784 ["Reconnaissance"] 561 ["Defense Evasion"] 29 ["Initial Access"] 4 ["Credential Access"] 1 Name: count, dtype: int64</pre>

CORRELATION ANALYSIS



COLUMNS TO DROP

Duplicate Columns

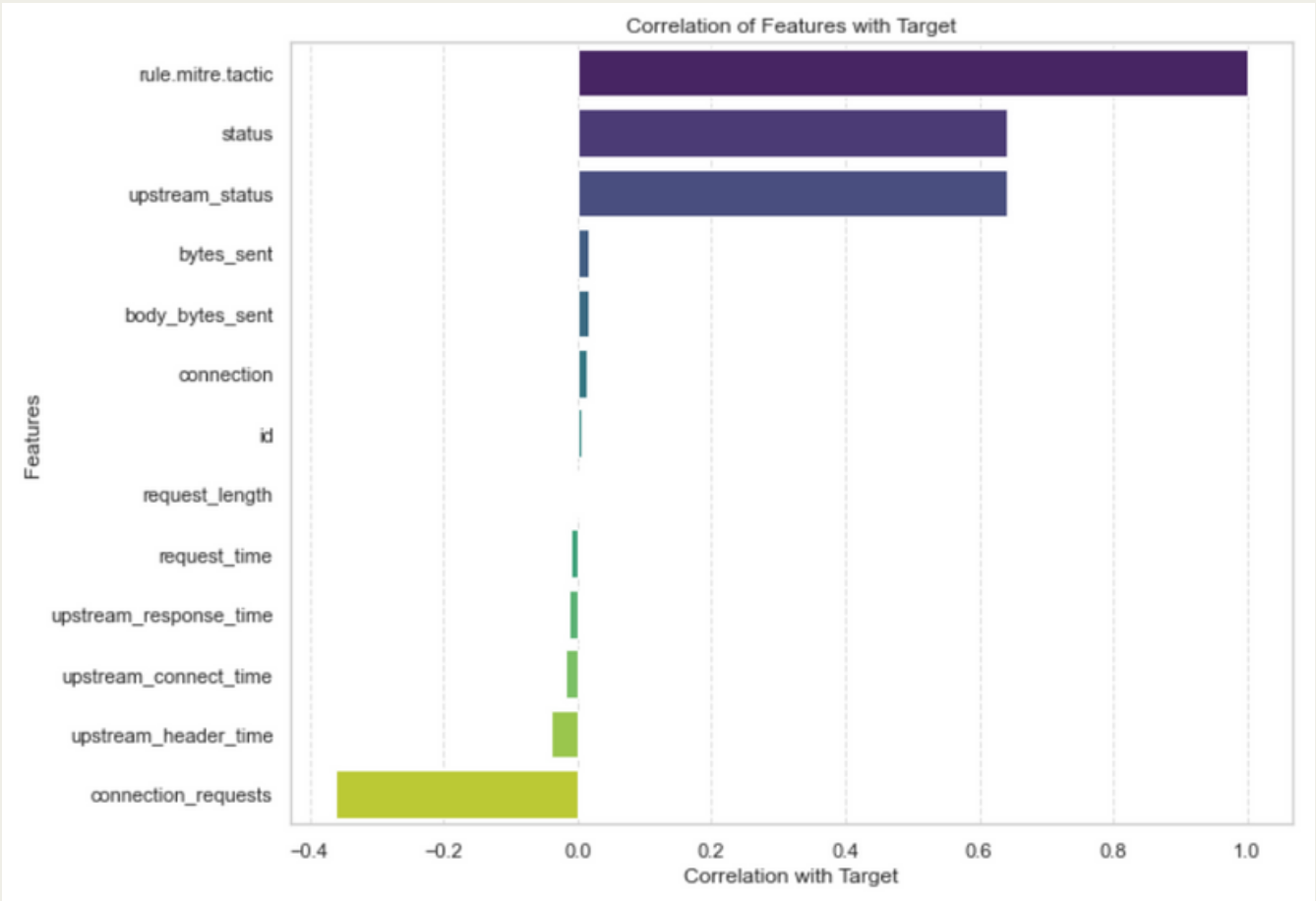
upstream_status	status
404	404
404	404
404	404
404	404
404	404
...	...
404	404
404	404
404	404
404	404
404	404

Only 1 Unique Value

```
1 train_set["hostname"].value_counts()
✓ 0.0s

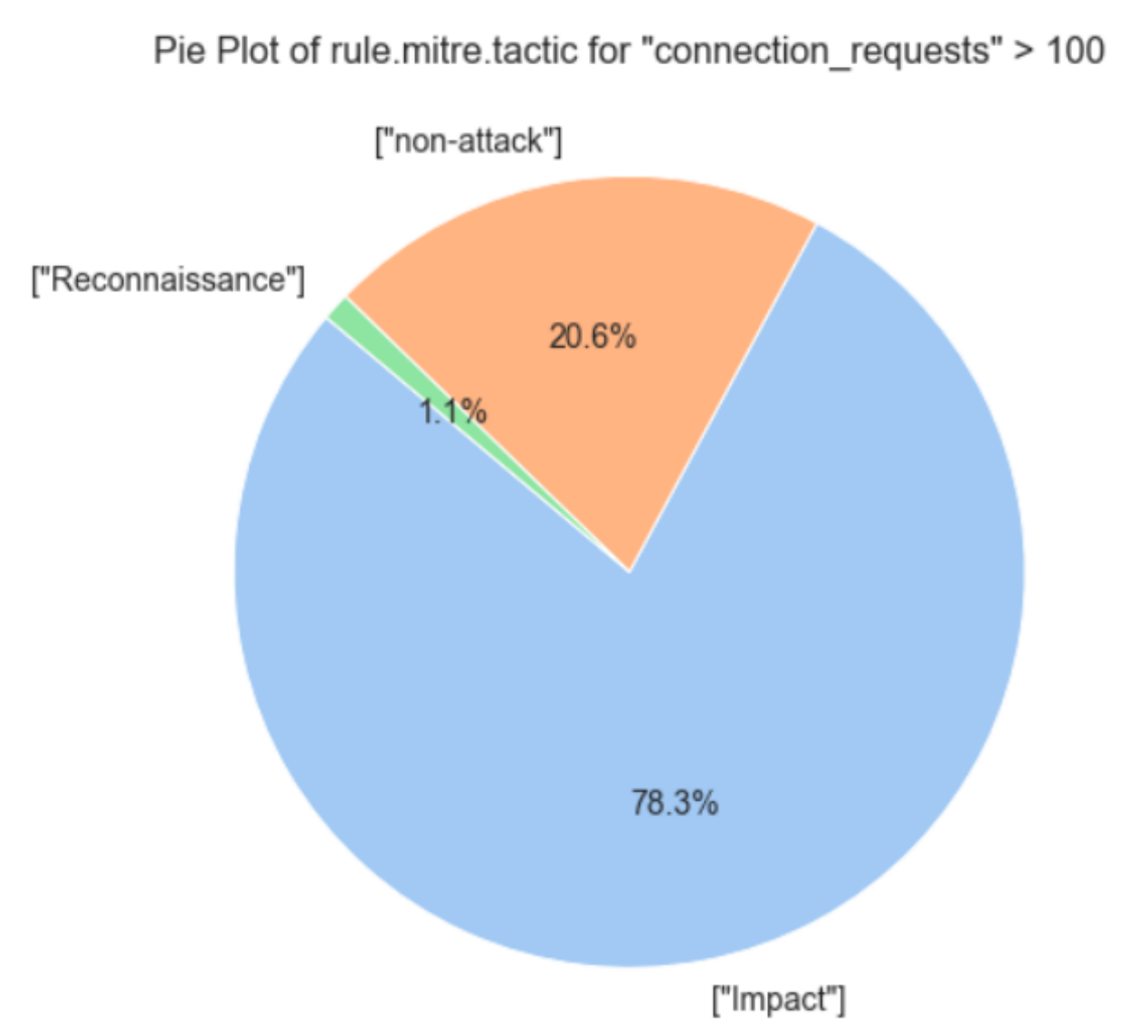
vm-director-nginx    54480
Name: hostname, dtype: int64
```

Low Correlation to Target



INTERESTING POINTS

Connection Requests > 100 : Favoured to Impact



```
1 train_set[train_set["status"] == 500][TARGET].value_counts()  
✓ 0.0s
```

```
["non-attack"]    116  
Name: rule.mitre.tactic, dtype: int64
```

```
1 train_set[train_set["status"] == 200][TARGET].value_counts()  
✓ 0.0s
```

```
["Impact"]          1772  
["Initial Access"]      4  
["Credential Access"]    1  
Name: rule.mitre.tactic, dtype: int64
```

Status Code 500 : Always Non-Attack
Status Code 200 : Always Cyber Attack

INTERESTING POINTS

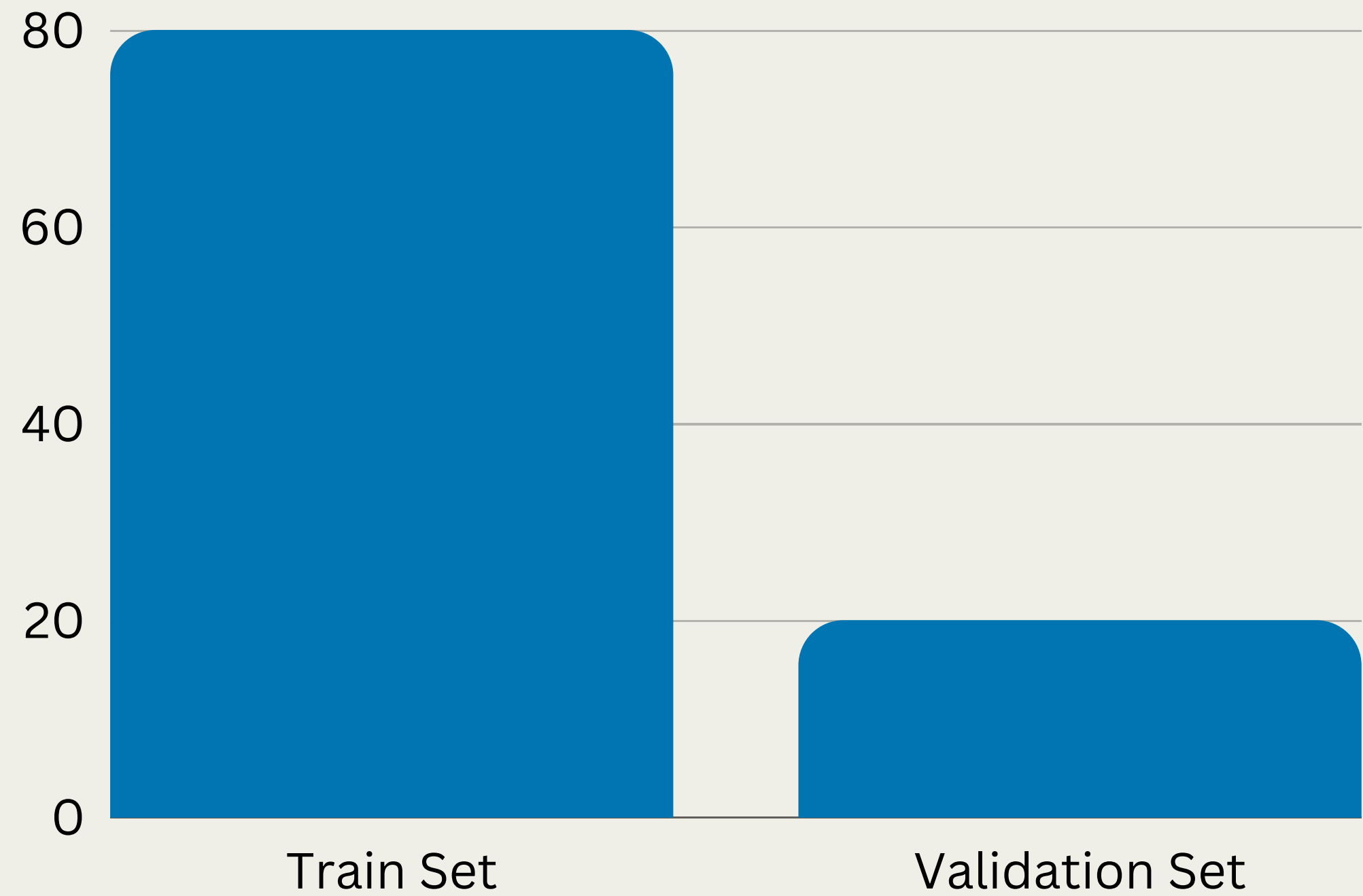
upstream_addr_10.45.184.59:8001	0.105317
upstream_addr_10.45.184.171:80	0.077581
...	...
upstream_addr_10.45.184.142:32401	-0.263361
upstream_addr_10.45.184.157:32401	-0.278160

Some upstream address can be handled uniquely due to its high correlation

Domain name can be extracted from HTTP Referer

-	47110
www.google.com	2000
https://mmd.notrealdomain.com/assets/827c7db8/css/theme.min.css	1505
https://vlm2.notrealdomain.com/my/	608
https://mmd.notrealdomain.com/	358
...	...
https://vlm2.notrealdomain.com/course/edit.php?id=14271	1
https://vlm2.notrealdomain.com/mod/quiz/review.php?attempt=590862&cmid=155453	1
https://halofilkom.notrealdomain.com/tickets.php?id=18573	1
https://vlm2.notrealdomain.com/course/view.php?id=14527	1
https://vlm2.notrealdomain.com/mod/assign/view.php?id=156151	1
Name: http_referer, Length: 977, dtype: int64	

VALIDATION TECHNIQUE



DESIGN OF EXPERIMENT (FEATURE ENGINEERING)

No	Design of Experiment	Evaluation
1	Handle imbalanced data by oversampling	Improve f1-score on validation set by 2%
2	Derive binary feature `request_above_100`	-0.45 correlation to target feature, Improved model performance on validation set by 1%
3	Simplify HTTP response status codes to its classes (Informational, Successful, Redirection, Client Error, Server Error)	Validation performance more stable , labels are better separated
4	Post processing by HTTP status code heuristic	Already handled by tree-based models
5	Different types of encoding on `upstream_address`	Rare encoding + One-hot encoding performs best

DESIGN OF EXPERIMENT (MODELLING)

No	Design of Experiment	Evaluation
1.	Base XGBoost	0.95 on Private LB
2.	Base LGBM	0.999 on Private LB
3.	Base CatBoost	Performs worse than XGBoost in validation set by 2%
4.	Base RandomForest	Underfits on training set
5.	Hyperparameter tuned LGBM with Optuna	Overfits on training set

```
Test set with 200 status code and login on wordpress or joomla: 0
```

**NOWHERE TO PREDICT CLASS
["CREDENTIAL ACCESS"]**

FINAL EVALUATION



LightGBM Validation Set Score

	precision	recall	f1-score	support
0	0.67	1.00	0.80	4
1	1.00	1.00	1.00	357
2	0.97	0.99	0.98	10206
3	1.00	1.00	1.00	1
4	0.22	0.08	0.11	328
accuracy			0.96	10896
macro avg	0.77	0.81	0.78	10896
weighted avg	0.95	0.96	0.96	10896

PUBLIC SCORE LEADERBOARD

Three Neurons		0.96241
Three Wise Monke		0.96170

PRIVATE SCORE LEADERBOARD

MALA		0.99942
Three Wise Monke		0.99934

ERROR ANALYSIS

Performs **well** on majority of data, **subpar** on “Defense Evasion” class, **poor** on “Initial Access” class.

CAUSE Lack of data supporting the target

Different feature importance caught by different model.

EXAMPLE XGBoost relies on “**status**” and “**request_method_POST**”, CatBoost highlights “**request_length**”, and LightGBM puts a lot of importance on “**connection**”

“Reconnaissance” class exploits **temporal** data

IMPORTANCE LightGBM **caught** this by putting emphasis on “connection” feature

FURTHER DEVELOPMENT

Modelling with data
partitioned by **heuristic**



Potentially produce more
robust model due to different
class distribution

Current model struggles on
predicting “**Reconnaissance**”
class



Incorporate more **time**-related
features

Incorporate **text features** as
attack description



SecureBERT model shows
86.7% accuracy on ATT&CK
attacking techniques

Thank you!

THREE WISE MONKE

